

Chahat Raj

PhD in Computer Science | George Mason University

🌐 chahatraj.github.io @ craj@gmu.edu 🎓 Google Scholar 📄 github.com/chahatraj 🐦 [chahatsaidit](https://twitter.com/chahatsaidit)

Research Interests

I work on NLP and Generative AI, specializing in Large Vision-Language Models, focusing on their harm evaluation & mitigation, interpretability, and alignment. My research explores socio-cultural biases, ethics, and fairness in multilingual and multimodal LLMs, aiming for responsible AI development. I have recently published at ACL, EMNLP, AIES, AAAI and ECIR.

Education

Present Aug 2022	George Mason University Ph.D. in Computer Science (GPA: 3.83) Advisors: Ziwei Zhu , Antonios Anastasopoulos	Fairfax, Virginia
Aug 2021 Aug 2019	Delhi Technological University Masters in Information Systems (Thesis) Advisor: Priyanka Meel	Delhi, India
May 2019 Aug 2015	Indira Gandhi Delhi Technical University for Women Bachelors in Computer Science & Engineering	Delhi, India

Experience

Aug 2026 May 2026	Amazon <i>Applied Scientist Intern</i> Mentor: Ren Pang , Manager: Zheng Qi > A parameter-efficient safety steering technique for Bedrock foundation models.	Seattle, WA
Aug 2025 May 2024	George Mason University, University of Washington <i>Graduate Research Assistant</i> Advisors: Ziwei Zhu , Antonios Anastasopoulos , Aylin Caliskan > Societal simulations, model unlearning, and cognitive alignment for bias mitigation.	Fairfax, VA
Sep 2024 May 2024	University of Washington Tech Policy Lab, The Information School <i>Visiting Scholar</i> Advisor: Aylin Caliskan > Bias, cognition, and mitigation in Vision Language Models.	Seattle, WA
Aug 2023 May 2023	George Mason University School of Computer Science <i>Graduate Research Assistant</i> Advisor: Ziwei Zhu > Fairness, interpretability, and robustness in transformer-based fake news detection.	Fairfax, VA
Jul 2022 Jun 2021	University of Technology Sydney School of Computer Science <i>Visiting Scholar</i> Advisor: Mukesh Prasad > Cyberbullying, AI for Aborigines, modeling disaster fundraising, and ML for clinical diagnosis.	Remote

Selected Publications

[full list here](#)

- [C.12] **VIGNETTE: Socially Grounded Bias Evaluation for Vision-Language Models** [PDF | Code]
[Chahat Raj](#), Bowen Wei, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
In Proceedings of the Association for Computational Linguistics: ACL 2026, San Diego, California, USA [ACL Main'26]
- [C.11] **Talent or Luck? Evaluating Attribution Bias in Large Language Models** [PDF | Code]
[Chahat Raj](#), Mahika Banerjee, Jinhao Pan, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
In Findings of the Association for Computational Linguistics: ACL 2026, San Diego, California, USA [ACL Findings'26]
- [C.10] **Knowing Bias, Doing Better: Mitigating Social Bias in LLMs via Know-Bias Neuron Enhancement** [PDF | Code]
Jinhao, [Chahat Raj](#), Anjishnu Mukherjee, Sina Mansouri, Bowen Wei, Shloka Yada, Ziwei Zhu
International Conference in Machine Learning: ICML 2026, Seoul, South Korea [ICML'26]
- [C.9] **Measuring South Asian Biases in Large Language Models** [PDF | Code]
Mamnuya Rinki, [Chahat Raj](#), Anjishnu Mukherjee, Ziwei Zhu
TrustNLP at ACL 2026 [TrustNLP'26]
- [C.8] **Discovering Bias Associations through Open-Ended LLM Generations** [PDF | Code]
Jinhao Pan, [Chahat Raj](#), Ziwei Zhu
In Proceedings of the AAAI Conference on Artificial Intelligence [AAAI'26]
- [C.7] **What's Not Said Still Hurts: A Description-Based Evaluation Framework for Measuring Social Bias in LLMs**
Jinhao Pan, [Chahat Raj](#), Ziyu Yao, Ziwei Zhu [PDF | Code]
In Findings of the Association for Computational Linguistics: EMNLP 2025, Suzhou, China [EMNLP Findings'25]

- [C.6] **Toward Inclusive Language Models: Sparsity-Driven Calibration for Systematic and Interpretable Mitigation of Social Biases in LLMs** [PDF | Code]
Prommy Sultana Hossain, [Chahat Raj](#), Ziwei Zhu, Jessica Lin, Emanuela Marasco
In Findings of the Association for Computational Linguistics: EMNLP 2025, Suzhou, China [EMNLP Findings'25]
- [C.5] **BiasDora: Exploring Hidden Biased Associations in Vision-Language Models** [PDF | Code]
[Chahat Raj](#), Anjishnu Mukherjee, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu
In Findings of the Association for Computational Linguistics: EMNLP 2024, Miami, Florida, USA [EMNLP Findings'24]
- [C.4] **Breaking Bias, Building Bridges: Evaluation and Mitigation of Social Biases in LLMs via Contact Hypothesis**
[Chahat Raj](#), Anjishnu Mukherjee, Aylin Caliskan, Antonios Anastasopoulos, Ziwei Zhu [PDF | Code]
Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society [AIES'24]
- [C.3] **Global Voices, Local Biases: Socio-cultural Prejudices across Languages** [PDF | Code]
[Chahat Raj](#)*, Anjishnu Mukherjee*, Ziwei Zhu, Antonios Anastasopoulos
In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, Singapore [EMNLP Main'23]
- [C.2] **SALSA: Saliency-Based Switching Attack for Adversarial Perturbations in Fake News Detection Models**
[Chahat Raj](#)*, Anjishnu Mukherjee*, Hemant Purohit, Antonios Anastasopoulos, Ziwei Zhu [PDF | Code]
European Conference on Information Retrieval [ECIR'24]
- [C.1] **True and Fair: Robust and Unbiased Fake News Detection via Interpretable Machine Learning** [PDF | Code]
[Chahat Raj](#), Anjishnu Mukherjee, Ziwei Zhu
Proceedings of the AAAI/ACM Conference on Artificial Intelligence, Ethics, and Society [AIES'23]

Selected Projects

- | | | |
|---|--|----------|
| > AI, Power & Society | Auditing Diffusion Models for Generational and Representational Errors | [Report] |
| > Computer Vision | Cultural Image Translation for Human Figures | [Report] |
| > AI: Ethics, Policy & Society | Implicit Association Test in Multimodal Generative Models | [Report] |

Teaching

- | | | |
|--|--|--------------------|
| > CS 484 Theory/Application Data Mining, GMU | <i>Inst.: Keren Zhou</i> | Spring'26 |
| > CS 584 Theory/Application Data Mining, GMU | <i>Inst.: Ziwei Zhu</i> | Fall'25 |
| > CS 108 Introduction to Computer Programming, GMU | <i>Inst.: Sara Lynn Gopalakrishna, Bella Yin</i> | Spring'24 |
| > CS 478 Natural Language Processing, GMU | <i>Inst.: Antonios Anastasopoulos</i> | Fall'23 |
| > COMP 502 Mathematical Foundations of Computing, GMU | <i>Inst.: William D Ellis</i> | Fall'23 |
| > CS 112 Introduction to Python Programming, GMU | <i>Inst.: Shahnaz Kamberi</i> | Fall'22, Spring'23 |

Student Mentorship

sub-mentoring with Ziwei, Antonis, Aylin

- | | |
|---|---|
| > Mamnuya Rinki [MS Thesis, GMU] | Mitigating South Asian Biases in multilingual open-ended LLM generations |
| > Angela Liu [TJ High School] | Assessing moral attributional biases in ambiguous contexts in LLMs |
| > Daksha Rajagopal [TJ High School] | Modeling normative ethics and cross-cultural moral reasoning in VLMs |
| > Yongxu Sun [MS, UW] | Inoculation Prompting by in-context learning to debias LLMs' decisions |
| > Mahika Banerjee [TJ High School] | Measuring reasoning and attribution biases across genders & nationalities |
| > Diwita Banerjee [MS, GMU] | Assessing cuisine biases; scientific figure explanation issues in VLMs |
| > Rinki, Sharanya, Aksh [Advanced NLP] | Intersectional biases in LLMs across Indo-Aryan languages |

Awards

- | | |
|--|--|
| > Outstanding Academic Award @ GMU, 2026 | Received an Outstanding Academic Achievement Award. |
| > Graduate Teaching Award @ GMU, 2026 | Received an Outstanding Graduate Teaching Assistant Award. |
| > Outstanding Reviewer Award @ EACL 2026 | Received an Outstanding Reviewer Award at EACL 2026. |
| > OpenAI Researcher Access Program Award 2025 | Funded \$1000 worth of OpenAI API credits. |
| > Travel Grant by AAAI/ACM AIES 2024 & 2023 | Funded \$2000 to attend AIES in San Jose and Montreal. |
| > CAHMP Fellowship @ GMU, 2024 | Summer research on social biases in question answering. |
| > Best Paper Award @ MASC-SLL 2024 | A Psychological View to Social Bias in LLMs. |
| > Summer Graduate Research Award @ GMU, 2023 | Explainability and Robustness of Debiasing. |
| > Research Excellence Award @ DTU 2023 & 2022 | Received three Commendable Research Awards with INR 150k. |
| > Graduate Scholarship @ DTU 2019 | Received INR 297k by AICTE for qualifying GATE. |

Program Committee

- | | |
|-------------------|---|
| > Reviewer | NeurIPS'26, ARR'23-26, CHI'26, AAAI'26, IASEAI'26, COLM'25-26, TrustNLP'24-26, LTEDI @ EACL'24, CSCW'24, NeurIPS'23, ACM TIST |
|-------------------|---|